LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Ch 11 Test (Set B): Electricity

Chapter / Topic: Electricity

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. In numericals, write the formula, substitute values with units, and give the final answer with its unit.

SECTION A

6 marks

Choose the correct option.

Q1. The SI unit of potential difference is the: **[1]**

(a) ampere (b) ohm
(c) coulomb (d) volt

Q2. The length of a wire is doubled and its area of cross-section is halved. Its resistivity: **[1]**

(a) doubles (b) stays the same
(c) halves (d) becomes four times

Q3. Two 4 Ω resistors are connected first in series and then in parallel. The ratio of the series resistance to the parallel resistance is: **[1]**

(a) 1 (b) 2
(c) 4 (d) 8

Q4. How much electrical energy does a 100 W bulb use in 10 hours? **[1]**

(a) 0.1 kWh (b) 1 kWh
(c) 10 kWh (d) 100 kWh

Q5. The filament of an electric bulb is made of: **[1]**

(a) copper (b) nichrome
(c) tungsten (d) aluminium

Q6. **Assertion (A):** Alloys are used in electric heating devices.
Reason (R): Alloys have a higher resistivity than pure metals and do not oxidise easily at high temperatures. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. A current of 0.5 A flows through a bulb for 10 minutes. How much charge flows through it? **[2]**

OR

How many electrons make up a charge of 1 coulomb? (Charge on an electron = 1.6×10^{-19} C.)

Q8. Why does the cord of an electric heater not glow, while its heating element does? **[2]**

SECTION C

6 marks

Q9. Resistors of 2 Ω, 3 Ω and 5 Ω are connected in series to a 10 V battery. **[3]**

(i) Find the total resistance. **[1]**

(ii) Find the current in the circuit. **[1]**

(iii) Find the potential difference across the 5 Ω resistor. **[1]**

Q10. A current of 2 A flows through a 50 Ω resistor for 1 minute. Calculate the heat produced. How would the heat change if the current were doubled? **[3]**

OR

A 1500 W electric heater works on a 220 V supply. Find the current it draws and its resistance. Can it be used safely on a circuit protected by a 5 A fuse?

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) State Ohm's law. Describe an experiment to verify it. **[3]**

(ii) What does the slope of a V–I graph (V on the y-axis, I on the x-axis) represent? Which of two such lines represents the larger resistance? **[2]**

OR

(a) Compare the power used by the 2 Ω resistor in each case: (i) a 6 V battery in series with 1 Ω and 2 Ω resistors; (ii) a 4 V battery in parallel with 12 Ω and 2 Ω resistors. [3]
(b) Why is a series arrangement not used in domestic circuits? [2]

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. A family uses five 10 W LED bulbs for 5 hours a day, two 75 W fans for 8 hours a day, and a refrigerator rated 150 W that effectively runs for 10 hours a day. Electricity costs ₹5 per unit (kWh). **[4]**

(i) How much energy do the LED bulbs use per day? **[1]**

(ii) What is the total energy used per day? **[1]**

(iii) Find the electricity bill for 30 days, and suggest one way to reduce it. **[2]**

~ All the best! ~